

Skills Summary

Subtracting Across a Zero with Regrouping

Use this reference while completing your worksheet or when studying.

1. Subtract across ONE zero (subtract-across-one-zero)

The rule: you cannot borrow from a 0 — a zero has nothing to give. Move *left* to the first digit that is not 0, take one from it, and hand the group back one place at a time.

1 hundred = 10 tens 1 ten = 10 ones

Say it as trading: “The tens are empty, so the hundreds give the tens ten. Now the tens can give the ones ten.”

Example: $403 - 167$

Step 1. The ones need more ($3 < 7$) and the tens digit is 0, so the tens cannot help — this is a trade across a zero, starting from the hundreds.

Step 2. $403 = 3$ hundreds, 10 tens, 3 ones $\rightarrow 3$ hundreds, 9 tens, 13 ones.

Step 3. $13 - 7 = 6$ ones; $9 - 6 = 3$ tens; $3 - 1 = 2$ hundreds $\Rightarrow$ **236**

Try it: $502 - 245$

check: **257**

2. Subtract across TWO zeros (subtract-across-two-zeros)

The rule: when the tens *and* the ones are both 0, the trade travels two places. Rewrite the whole number before you subtract anything:

$600 = 5$ hundreds + 9 tens + 10 ones

The digits 5, 9, 10 are the pattern for every “hundreds minus something” problem: one less hundred, nine tens, ten ones.

Example: $600 - 274$

Step 1. Ones and tens are both 0, so nothing can be borrowed until the hundreds break — rewrite 600 first, then subtract once.

Step 2. $600 = 5$ hundreds, 9 tens, 10 ones.

Step 3. $10 - 4 = 6$; $9 - 7 = 2$; $5 - 2 = 3 \Rightarrow$ **326**

Try it: $900 - 358$

check: **542**

3. Estimate the difference first (estimate-the-difference)

The rule: round *each* number to the same place, then subtract the rounded numbers. Rounding to the nearest hundred: look at the tens digit — 5 or more rounds up, less than 5 rounds down.

An estimate is a *safety net*: if your exact answer is far from the estimate, you dropped a trade somewhere.

Example: Estimate $497 - 212$ to the nearest hundred.

Step 1. This is an estimate, not the exact answer, so round both numbers before subtracting instead of regrouping.

Step 2. $497 \rightarrow 500$ (tens digit 9); $212 \rightarrow 200$ (tens digit 1).

Step 3. $500 - 200 = 300$, so the exact answer should be near 300.

Try it: Estimate $803 - 396$ to the nearest hundred.

check: 407

4. Check by adding (check-by-adding)

The rule: addition undoes subtraction.

If $\text{top} - \text{bottom} = \text{answer}$, then $\text{answer} + \text{bottom} = \text{top}$.

Add your answer to the number you subtracted. If you get the number you started with, the trades were right.

Example: Is $405 - 218 = 187$ correct?

Step 1. Checking a difference means running it backwards, so add the answer to the number that was subtracted.

Step 2. $187 + 218$: $7 + 8 = 15$ (write 5, carry 1); $8 + 1 + 1 = 10$ (write 0, carry 1); $1 + 2 + 1 = 4$.

Step 3. The check gives 405, the number we started from, so 187 is correct.

Try it: Check $300 - 126 = 174$ by adding $174 + 126$.

check: 300

(!) Watch out: Never subtract the smaller digit from the larger one just because it is easier. In $903 - 458$ the ones column is $3 - 8$, *not* $8 - 3$. Turning it around gives 555 instead of the correct 445.